Seat No: ____________ Enrollment No: __________________

PARUL UNIVERSITY

FACULTY OF ENGINEERING & TECHNOLOGY

B.Tech. Winter 2021 - 22 Examination

Semester: 5 **Date: 20-10-2021**
Subject Code: 03105303 **Time: 10:30 am to 01:00 pm**
Subject Name: Theory of Computation **Total Marks: 60**

Instructions:

1. All questions are compulsory.
2. Figures to the right indicate full marks.
3. Make suitable assumptions wherever necessary.
4. Start new question on new page.

Q.1 Objective Type Questions (All are compulsory) (Each of one mark) **(15)**

1.Deterministic and Non deterministic Push down automata are
A) equally powerful
B) same thing but just different convention
C) always convertible form Deterministic PDA to Non deterministic PDA
D) none of the above

2.If n count of input's are given to Mealy Machine than outputs depends on
A) present state
B) present state and present input
C) present input
D) none of the above

3. Regular expression for the language accepting any binary string with odd count of 1 is
A) 10*(0*10*10*)*
B) (0*10*10*)*1 0*
C) (0*10*10*)*0*10*
D) 0*1(0*10*10*)*

4. Finite Automata can parse only Regular language not other not other languages because
A) It has restriction over number of states
B) It has limited number of resources
C) It does not have memory
D) All of the above

5. NFA with epsilon and NFA without epsilon transition contains
A) Equal number of states
B) Cannot be defined about the number of states
C) NFA with epsilon will have always more number of states
D) NFA without epsilon will have always more number of states

6. ___________ Turing machine is a type of Turing machine that simulates an arbitrary Turing machine on arbitrary input.

7.___________ language is accepted by turing machine.

8.___________ language is accepted by pushdown automata.

9. __________ language is called type 2 grammar.

10. Arden's theorem can be used to construct ____________ for given equivalent Finite Automata.

11. The power of NFA without epsilon. NFA with epsilon and DFA is __________

12. L1 and L2 are regular language then L1 intersection L2 is _____________

13. For a given regular language if we draw DFA and NFA than possibly more number states will be in ________

14. Minimization always decreases the number of states in DFA. True / False

15. __________ theorem says a language is regular, i.e. it can be defined by a regular expression, if and only if it is recognized by finite automata.

Q.2 Answer the following questions. (Attempt any three) **(15)**

A) State and brief principal of mathematical induction with suitable example

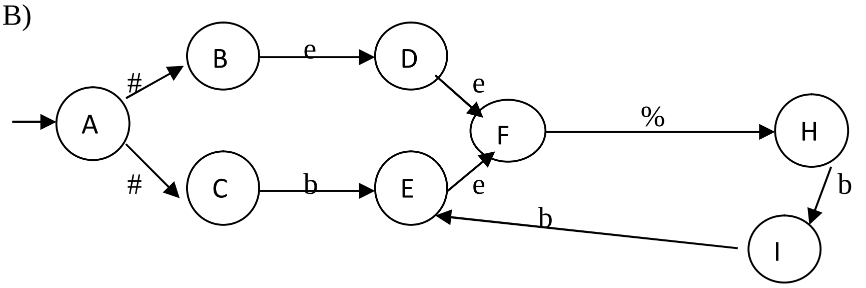
B) Justify proof by contradiction with suitable example

C) Define ambiguous grammar , left recursive grammar, Non deterministic grammar and Sentential form

D) State and explain pumping lemma for regular Language with suitable example.

Q.3 A) Define Turing machine. Explain all the 7-tuple of TM with a suitable example. **(07)**

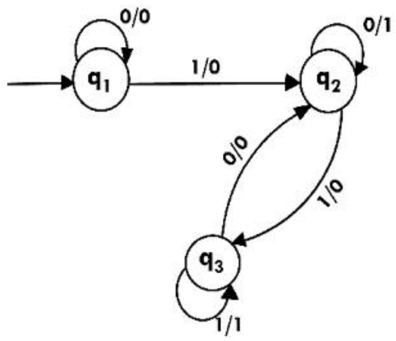
B) **(08)**



Remove Epsilon transition and then convert it into DFA. Here #, b and % are symbols and 'e' stands for epsilon. State 'I' is the final state and state 'A' is start symbol.

OR

B) Convert the below mealy machine to equivalent Moore machine. **(08)**



Q.4 A) List and state notation of all the tuple of PDA and construct PDA that accepts the language $L=\{a^n b^{n+m} c^m$ where $n \geq 1, m \geq 1\}$ **(07)**

OR

A) Construct PDA that accepts the language $L=\{a^n b^{3n} c^m\ d^{m+m}$ where $n \geq 1, m \geq 1\}$ **(07)**

B) Construct the Regular grammar for palindrome with alphabet (a,b) and then convert it into linear grammar **(08)**